

# Prompt2Model: Language-Guided Vision Model Factory

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## Abstract

*Prompt2Model turns a free-form request into a runnable vision pipeline. The current prototype covers typed prompt parsing, label resolution, augmentation selection, lightweight model choice for classification and COCO-style detection, smoke-test training, and ONNX export. On a 12-prompt rubric, task and label extraction are 91.7% exact-match, while priority and environment tags are 100%; the label resolver is 100% on an 8-case synonym check. For real-data evaluation, we downloaded Beans and CIFAR-10. On Beans, the prompt-guided low-light recipe improves corrupted-test accuracy from 44.5% to 62.5%; on CIFAR-10, the prompt-guided motion-blur recipe improves blur accuracy from 38.7% to 66.5%. A Beans ablation also shows that pretrained initialization matters and that EfficientNet-B0 outperforms MobileNetV3-Small under the same recipe.*

## 1. Introduction

Building a vision model still involves many small engineering choices: task type, class names, augmentation, backbone, and export settings. Prompt2Model uses the prompt as the interface for those choices. In the current system, phrases such as “prioritize speed”, “low light”, or “motion blur” are translated into an explicit training recipe.

This report captures the week-4 integrated prototype plus real-data extensions. The aim is to show that the pieces work together end to end, with evidence beyond toy smoke outputs.

## 2. Method

The pipeline has three pieces. **Parsing:** a typed schema stores task type, requested labels, deployment constraints, and environment tags, and a deterministic parser fills that schema from prompt text. **Label grounding and model choice:** requested labels are matched to dataset classes with a lexical resolver today, while the CLIP path is already wired for later use [3]; the parsed priority signal selects lightweight backbones, currently MobileNetV3 for classification [1] and SSDLite320-MobileNetV3 for detection [1, 2]. **Training and export:** the system loads ImageFolder-like classifi-

Table 1. Real benchmark summary across two datasets. Both comparisons keep the backbone fixed and change only the prompt-activated augmentation recipe.

| Dataset  | Split     | Fixed | Guided | $\Delta$ |
|----------|-----------|-------|--------|----------|
| Beans    | clean     | 94.5% | 90.6%  | -3.9     |
| Beans    | low-light | 44.5% | 62.5%  | +18.0    |
| CIFAR-10 | clean     | 78.5% | 79.0%  | +0.5     |
| CIFAR-10 | blur      | 38.7% | 66.5%  | +27.7    |

cation data or COCO-style detection annotations, trains a model, evaluates it, and exports an ONNX artifact with embedded metadata. In the current implementation, “prioritize speed” selects MobileNetV3-Small, “low light” enables brightness/contrast plus blur, and “motion blur” enables blur augmentation during training.

## 3. Experimental Settings

To verify integration, we combine repository-level checks with two real-data benchmarks. The validation tracks are a 12-prompt parsing rubric, an 8-case synonym benchmark, a Beans low-light study, a CIFAR-10 motion-blur study, and smoke pipelines for classification and COCO-style detection export. For Beans, we used the official Hugging Face train/validation/test splits (1034/133/128 images). For CIFAR-10, we used a balanced 10k-train/2k-val subset from the standard training split plus the full official test set. Both studies use ImageNet-initialized MobileNetV3-Small and compare a fixed recipe against a prompt-guided corruption-specific recipe. Training ran on Apple Silicon MPS; latency and ONNX measurements were collected on CPU.

## 4. Preliminary Results

Figure 1 shows the Beans benchmark. The main result is a clear robustness tradeoff: the prompt-guided low-light recipe improves corrupted-test accuracy from 44.5% to 62.5%, a gain of 18.0 points, while clean accuracy changes from 94.5% to 90.6%. The second benchmark shows the same pattern on CIFAR-10: the motion-blur recipe improves blur accuracy from 38.7% to 66.5% with almost no clean penalty (78.5% to 79.0%). Table 1 summarizes both benchmarks.

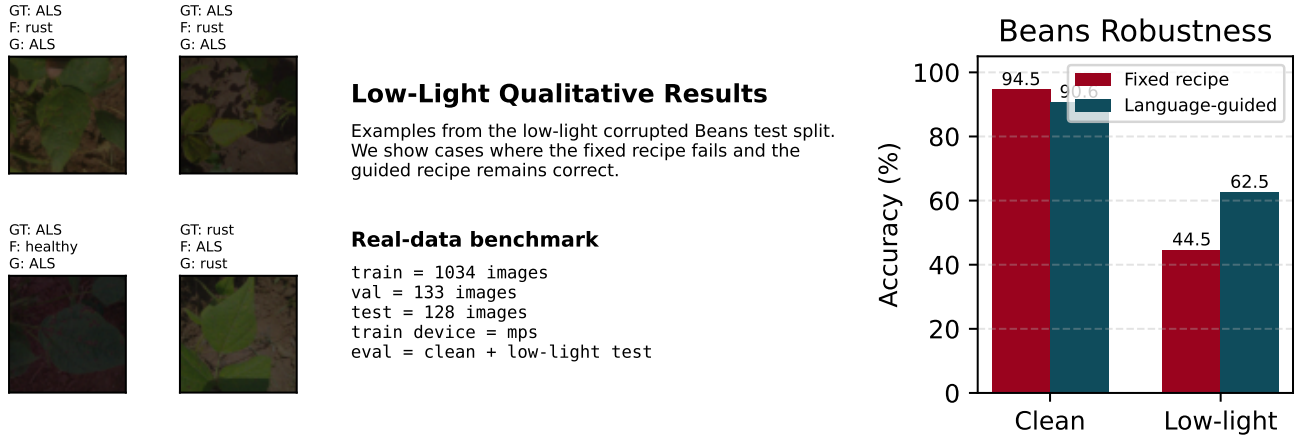


Figure 1. Beans robustness results from the implemented prototype. Left: fixed-recipe failures corrected by the guided recipe. Right: clean and corrupted test accuracy.

Table 2. Beans ablation under the same training budget.

| Config           | Init     | Clean | Low-light |
|------------------|----------|-------|-----------|
| Fixed MNV3-S     | ImageNet | 94.5% | 44.5%     |
| Guided MNV3-S    | ImageNet | 90.6% | 62.5%     |
| Guided MNV3-S    | scratch  | 33.6% | 33.6%     |
| Guided EffNet-B0 | ImageNet | 95.3% | 66.4%     |

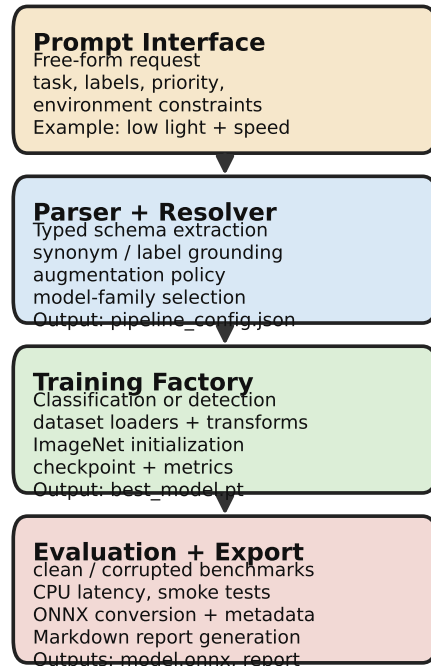
Table 2 shows the Beans ablation. Pretrained initialization matters: the guided MobileNetV3-Small model trained from scratch collapses to 33.6% on both clean and low-light evaluation. Backbone choice also matters after parsing: guided EfficientNet-B0 reaches 95.3% clean and 66.4% low-light accuracy, improving on guided MobileNetV3-Small. The parser and deployment stack are stable enough for these experiments: task and label extraction are both 91.7% exact-match, label resolution is 100% on 8 curated synonym cases, both smoke pipelines export to ONNX successfully, and CPU latency is 10.2 ms for MobileNetV3-Small and 28.7 ms for SSDLite320.

## 5. Pipeline Architecture

Figure 2 makes the system boundary explicit. The key claim is not only that prompts select augmentations, but that they control a full path from typed configuration to exported artifact. In the repo, those handoffs appear as `pipeline_config.json`, `best_model.pt`, `evaluation_report.md`, and `model.onnx`. The main missing piece is still real detection evaluation.

## References

[1] Andrew Howard, Mark Sandler, Grace Chu, Liang-Chieh Chen, Bo Chen, Mingxing Tan, Weijun Wang, Yukun Zhu, Ruoming Pang, Vijay Vasudevan, Quoc V Le, and Hartwig Adam. Searching for mobilenetv3. In *Proceedings*



Prompt-conditioned handoff chain implemented end to end.

Figure 2. Implemented Prompt2Model handoff chain.

of the IEEE/CVF International Conference on Computer Vision, 2019.

[2] Wei Liu, Dragomir Anguelov, Dumitru Erhan, Christian Szegedy, Scott Reed, Cheng-Yang Fu, and Alexander C. Berg. SSD: Single shot multibox detector. In *Proceedings of the European Conference on Computer Vision*, 2016.

[3] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the International Conference on Machine Learning*, 2021.